



Cairo University - Faculty of Engineering

CMPS450

Pattern Recognition & Artificial Neural Networks

Industrial Machine Fault Recognition Using Audio

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Abstract

This report presents a comprehensive methodology for diagnosing mechanical faults in rotating machinery utilizing acoustic signals. To address the inherent noise and high variance of real-world audio, a robust preprocessing pipeline was developed, featuring dynamic decibel-based trimming, 4 kHz low-pass filtering, and sliding window segmentation coupled with statistical quality assurance gates. The study evaluates two distinct predictive maintenance frameworks: an Edge Machine Learning approach utilizing statistical feature extraction with a Random Forest classifier, and a Deep Learning architecture employing Mel-spectrograms processed by a ResNet18 **Convolutional Neural Network (CNN)**.

Initial baseline experiments utilizing the **Random Forest** classifier revealed a severe accuracy paradox caused by an extreme class imbalance, resulting in the model missing approximately 74% to 79% of actual mechanical faults. While the application of algorithmic class weighting successfully doubled the fault detection recall (ranging from 49% to 61%), the traditional machine learning approach ultimately hit a performance ceiling. Consequently, the research transitioned to the advanced ResNet18 CNN framework. By successfully leveraging Focal Loss to penalize minority misclassifications, SpecAugment for robust noise injection, and an optimal majority-voting inference strategy, the deep learning architecture overcame the dataset's limitations to establish a highly reliable, enterprise-grade predictive maintenance solution.

Pre Processing

Data Exploration & Initial Assessment

In the preprocessing phase checking how raw data appears like was the first The preprocessing phase commenced with an exploratory data analysis to evaluate the raw audio signals. Recognizing that real-world acoustic data is inherently noisy and inconsistent, proper transformation into a structured, uniform format is a vital prerequisite for reliable machine learning model training (Nichols, 2026). Initial visualizations of the raw waveforms (see Figure 1) revealed distinct silent phases and unique frequency distributions across the different machinery classes. To systematically analyze these profiles, random samples of 100 audio files from each machine type spanning both normal and abnormal operational states were isolated and visualized individually.

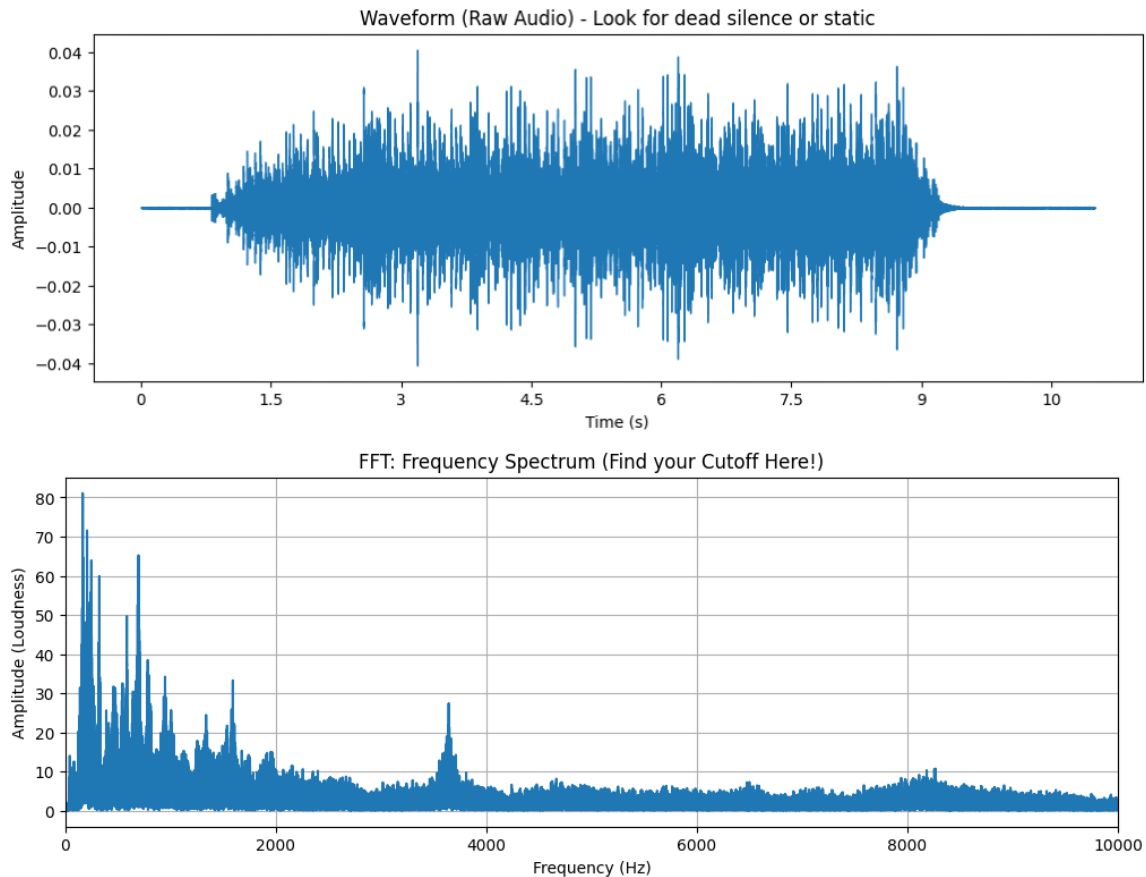


Fig 1 Sample audio data

Pipeline:

Trimming & Noise

Following the exploratory phase, a trimming protocol was implemented to eliminate leading and trailing silences. Initial iterations utilized a hardcoded temporal threshold; however, the high variance in recording lengths ((Fig 2) ranging up to 11.0 seconds) resulted in the unintended truncation of critical mechanical transients. To mitigate this, a dynamic audio signal processing algorithm (Dynamic Distribution Analyzer) was adopted. This algorithm dynamically trims silence based on a relative decibel threshold compared to the peak amplitude of the signal. Empirical testing demonstrated that a 26 dB threshold yielded the most optimal retention rate, effectively isolating active machine noise with a minimal 5% data loss.

Subsequent frequency domain analysis (Figures 3, 4, and 5) indicated that spectral energies above 4 kHz predominantly consisted of background noise and lacked diagnostic significance for mechanical faults in this specific machinery. Consequently, a low-pass filter was applied to attenuate these high-frequency artifacts. Following this noise reduction step, peak normalization was executed to rescale the amplitude of each audio file to a consistent maximum of 1.0, ensuring a uniform scale across the entire dataset.

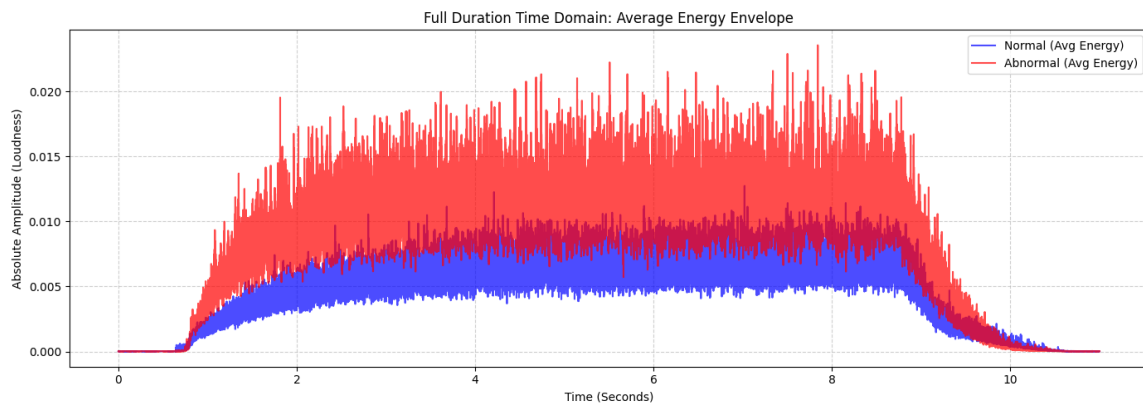


Fig 2 Example Data-Time Distribution Machine 1

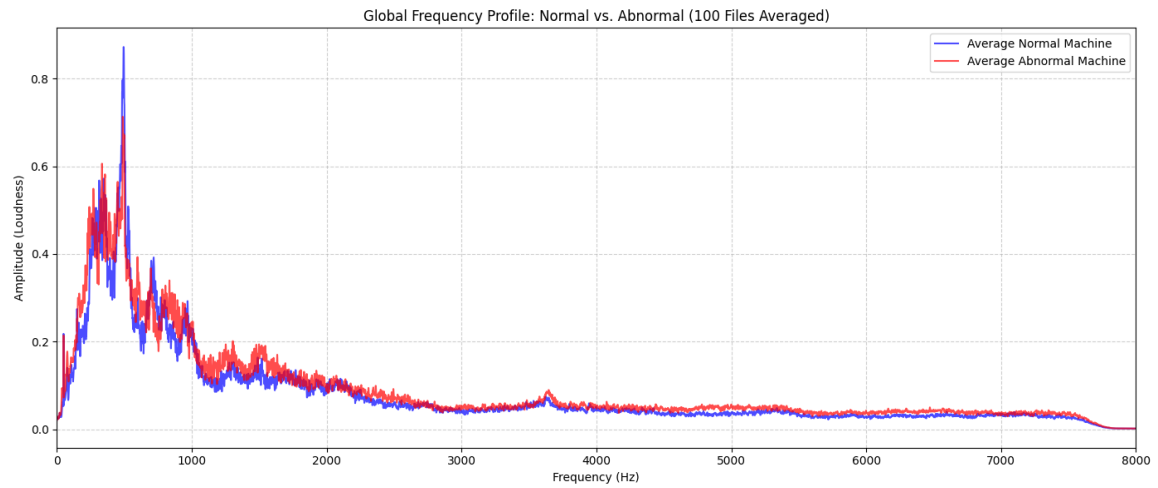


Fig 3 Machine 1 Frequency Domain

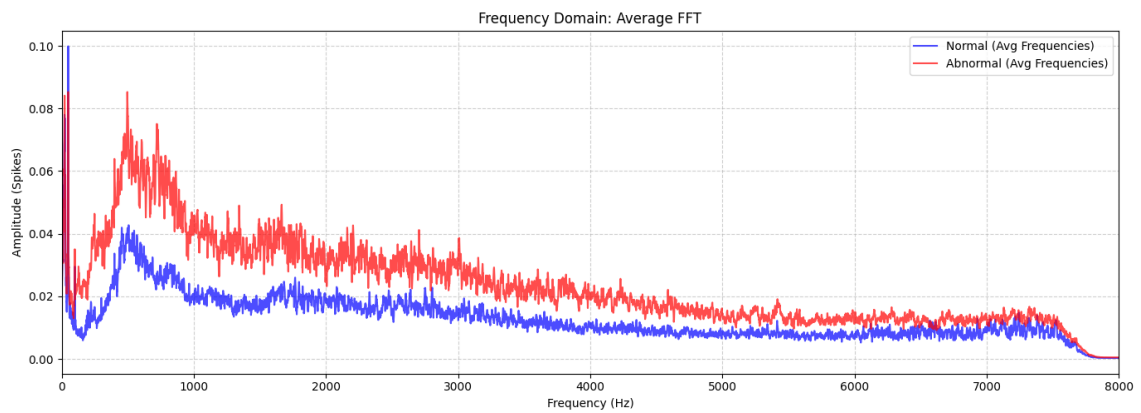


Fig 4 Machine 2 Frequency Domain

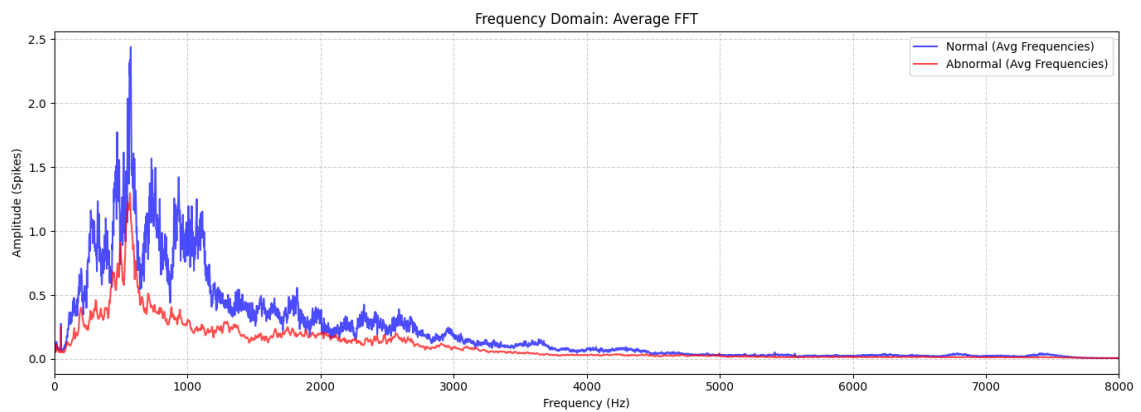


Fig 5 Machine 3 Frequency Domain

Segmentation Via Sliding Windows:

To train our parallel Convolutional Neural Network (CNN) and Edge Machine Learning models, the input data required uniform temporal dimensions. Rather than cropping a fixed timeframe from each file which risks discarding critical fault signatures the audio was segmented into continuous, overlapping chunks. The sliding window technique (GeeksforGeeks, n.d.) was employed to extract 0.5-second chunks with a 0.25-second overlap. This overlapping segmentation guarantees that continuous mechanical anomalies are captured across window boundaries without data loss, facilitating robust feature extraction for CNN-based predictive maintenance frameworks (Hafiz et al., 2025).

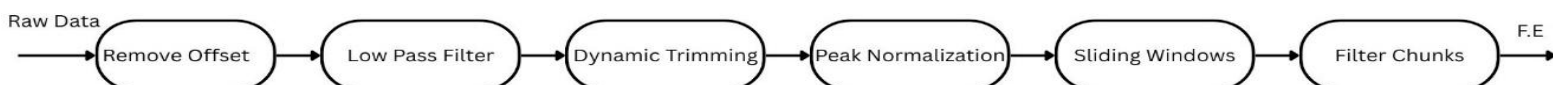
Quality Assurance Filtering Gates process:

Not all extracted chunks contained useful mechanical data; therefore, two statistical filtering gates were implemented to automatically discard low-quality segments (Mohammed et al., 2020). The first gate evaluated **the Coefficient of Variation (CV)** defined as the standard deviation divided by the mean to measure energy dispersion. A **CV** threshold of 1.2 was established to reject chunks containing irregular energy behaviors such as **ramp-ups** (quiet starts, loud ends), **ramp-downs**, **sudden dropouts**, or **isolated impacts**. The second gate utilized **Root Mean Square (RMS) energy**. Chunks with an **RMS** lower than **30%** of the parent file's peak RMS were rejected, successfully filtering out near-silent segments that had bypassed the initial trimming phase.

Parameter Optimization & System Trade-offs:

The specific thresholds for chunking and filtering were established through iterative empirical testing. Initial experiments using 2.0-second chunks with a 1.0-second overlap were suitable for Machines 1 and 3 due to their stable frequency profiles. However, these strict parameters inadvertently eliminated the critical, high-energy fault spikes inherent to Machine 2, leading to severe data degradation for that class. To balance preprocessing recall against model precision, data survival was prioritized. The combination of a 0.5-second chunk size, a high **CV** tolerance (1.2), and an aggressive **RMS** threshold (0.30) preserved the mechanical transients of Machine 2 while maintaining a high signal-to-noise ratio. This robust preprocessing pipeline ensures that the data is highly optimized for lightweight Edge ML deployments, aligning with recent advancements in **real-time fault detection for rotating machinery** (Shabita et al., 2023). Furthermore, while pre-emphasis filtering was evaluated to artificially boost high-frequency components, it was ultimately discarded as it distorted the specific frequency bands critical to our diagnostic requirements.

Fig 6 Full Preprocessing pipeline



Results:

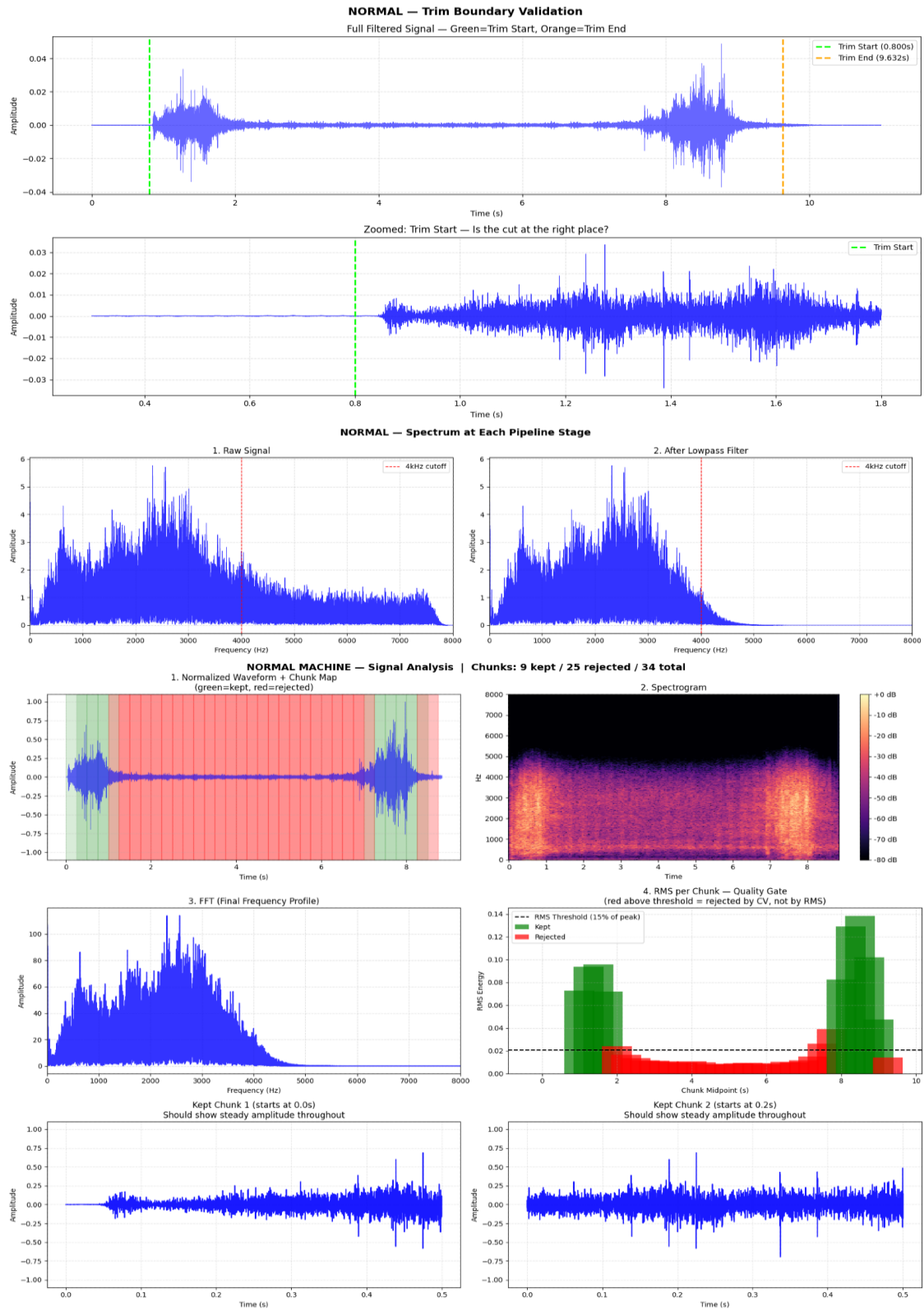


Fig 7 Machine 2 sample preprocessing

Models:

Three papers were mainly used in our approach that we say are similar to our project so their approach along with our tweaks was adopted.

Deep Learning:

Recent literature demonstrates that the combination of Mel-spectrograms and Convolutional Neural Networks (CNNs) yields superior diagnostic performance in industrial predictive maintenance applications (Hafiz et al., 2025). Consequently, we adopted this framework over traditional Mel-Frequency Cepstral Coefficients (MFCCs). While MFCCs are highly effective for speech recognition, they undergo a Discrete Cosine Transform (DCT) that heavily compresses the signal and intentionally discards spatial relationships between adjacent frequency bands. Because CNNs rely heavily on learning localized 2D spatial correlations, the uncompressed, highly detailed feature maps provided by Mel-spectrograms are significantly more suitable for capturing the nuanced, non-linear signatures of mechanical faults.

The transformation from raw audio to Mel-spectrograms was calibrated using standard industry practices optimized for mechanical noise:

- **Target Sample Rate (TARGET_SR = 16000):** Captures frequencies up to 8 kHz (per the Nyquist theorem), which adequately encompasses the critical mechanical acoustic ranges while remaining computationally efficient.
- **Number of Mel Bands (N_MELS = 128):** Provides a high-resolution frequency axis, allowing the model to distinguish fine-grained spectral differences.
- **Fast Fourier Transform Window (N_FFT = 1024):** Dictates the frequency resolution. A window of 1024 strikes an optimal balance, providing sufficient frequency detail without excessively blurring temporal events.
- **Hop Length (HOP_LENGTH = 512):** Defines the sliding window step size, resulting in a 50% overlap that ensures smooth temporal continuity across the spectrogram.

Network Architecture and Computational Environment

For the image processing of the generated Mel-spectrograms, ResNet18 was selected as the backbone architecture. It is widely regarded as an industry standard for spectrogram classification due to its residual connections, which effectively mitigate the vanishing gradient problem while maintaining a lightweight parameter footprint suitable for rapid inference.

The model was implemented utilizing the PyTorch ecosystem specifically the torch, torchaudio, and torchvision libraries. This allowed for native GPU acceleration and parallel processing. Training was executed within a Kaggle environment to leverage advanced GPU capabilities. Furthermore, data loading bottlenecks were minimized by setting the num_workers parameter to 4, which empirical testing confirmed as a safe and optimal configuration for continuous data streaming to the GPU.

Addressing Class Imbalance: Focal Loss

To counteract the severe class imbalance in the dataset (where "Normal" data heavily outweighed "Abnormal" data), the standard Cross-Entropy loss was replaced with a **Focal Loss** function. While traditional methods, such as Class Weights, attempt to fix imbalance by applying rigid, static mathematical multipliers to minority classes, Focal Loss acts dynamically. It actively scales down the loss assigned to easy, well-classified examples (i.e., the abundant Normal data) and focuses the model's penalization on hard-to-predict, underrepresented examples (the Abnormal faults). This ensured the CNN learned the complex fault boundaries rather than blindly guessing the majority class to achieve superficial accuracy.

Data Augmentation

To ensure the model learned core distinguishable mechanical features rather than overfitting to specific background noises, **SpecAugment** was applied exclusively to the training set. SpecAugment acts as a noise injection technique by masking random blocks of continuous frequencies and time steps. Through trial and error, the parameters were tuned to frequency_mask = 10 and time_mask = 20.

During initial monitoring, the training accuracy appeared substantially lower than the validation accuracy. This discrepancy is an expected artifact of aggressive SpecAugment: the model is evaluated on heavily distorted "masked" data during training, but evaluated on clean, unaltered data during validation. These masking parameters were iteratively tuned until a healthy, generalizable ratio was achieved, ensuring the final system is robust and ready to classify incoming real-world data of varying recording qualities.

Training Dynamics and Early Stopping Logic

The training loop was governed by the following hyperparameters: a BATCH_SIZE of 64, a LEARNING_RATE of 0.001, and a maximum epoch limit set aggressively high (EPOCHS = 30). The high epoch count was intentionally selected because training duration was actively managed by a custom **Early Stopping** convergence logic.

Crucially, the early stopping mechanism monitored the **Validation Macro F1-Score** rather than overall accuracy. Accuracy is a highly misleading metric in imbalanced datasets, whereas the Macro F1-score heavily penalizes the misclassification of small minority classes. The logic required a Minimum Delta (MIN_DELTA = 0.5%) improvement. If the model achieved this, the state dictionary was saved (best_machine_model.pth). If the model failed to yield meaningful improvement for a Patience of 4 epochs (PATIENCE = 4), training was halted to conserve computational resources and prevent overfitting.

Iterative Prototyping on Toy Dataset

To efficiently tune the aforementioned parameters and validate the model's convergence trajectory, initial prototyping was conducted on a "toy dataset." This subset comprised exactly 11,600 audio chunks per class, representing approximately 5% of the total dataset. Three distinct experimental trials guided our final configuration:

- **Trial 1:** Utilized accuracy as the stopping metric alongside highly aggressive SpecAugment masking parameters.
 - *Epoch 5 Results:* Train Loss: 1.0818 | Train Acc: 50.24% | Val Loss: 0.1703 | Val Acc: 93.53%.
(The massive gap highlighted the extreme difficulty of the augmented training data compared to the clean validation data).
- **Trial 2:** Noise injection masking parameters were tuned down to more reasonable levels.
 - *Epoch 10 Results:* Train Loss: 0.4582 | Train Acc: 79.45% | Val Loss: 0.0661 | Val Acc: 97.63%.
(Convergence improved significantly).
- **Trial 3:** Transitioned from Accuracy to Macro F1-Score as the primary evaluation metric and integrated Focal Loss to penalize minority misclassifications.
 - *Epoch 9 Results:* Train Focal Loss: 0.2966 | Val Focal Loss: 0.0389 | Val Macro F1: 95.04%.

The success of Trial 3 confirmed the architecture's stability and metric reliability, justifying the deployment of this exact configuration onto the full dataset.

Inference Strategy and Chunk Aggregation

During the deployment and inference phase (infer.py), the CNN processes the extracted Mel-spectrograms and generates independent predictions for each overlapping 0.5-second chunk. However, the ultimate diagnostic objective is to determine the holistic operational state of the entire continuous audio recording.

To synthesize these localized, chunk-level predictions into a definitive parent-file classification, a majority voting mechanism was implemented. Under this paradigm, if 50% or more of the constituent chunks within a given audio file are classified as "Abnormal," the entire file is subsequently flagged as faulty. Prior to finalizing this 50% threshold, rigorous empirical threshold tuning was conducted to investigate whether specific machinery classes required custom fault density tolerances. The empirical results definitively validated that custom, class-specific fault densities were unnecessary; the standard 50% majority voting approach proved highly robust and optimal for continuous file-level classification across all machine types.

Final Results:

Following the hyperparameter optimization and the integration of Focal Loss and SpecAugment, the final ResNet18 Convolutional Neural Network was evaluated on the unseen test set (comprising over 272,000 chunks). The empirical results conclusively demonstrate the superiority of the Deep Learning architecture over the Edge Machine Learning baseline.

Whereas the statistical Random Forest baseline struggled with the accuracy-recall trade-off maxing out at a 61% fault detection recall and 77% overall accuracy the CNN architecture effectively solved the class imbalance problem. The final model achieved an extraordinary **overall accuracy of 99.75%** and a **macro-average F1-score of 99.18%**.

A detailed breakdown of the classification report highlights the system's robust diagnostic capabilities across all machinery:

- **Machine 3:** Achieved perfect classification (1.0000 Precision, Recall, and F1-score) for both normal and abnormal operational states.
- **Machine 1:** Demonstrated near-perfect fault detection, resolving the abnormal state with a 99.32% recall and an F1-score of 0.9923.
- **Machine 2:** Previously identified as the most challenging class due to its highly transient mechanical spikes, Machine 2 saw a massive leap in fault detection. The CNN achieved an abnormal recall of 96.60% and an F1-score of 0.9670.

Edge Machine Learning:

Feature Extraction Methodology

For Track B, the fault detection framework was formulated as a supervised multi-class classification problem. The extraction pipeline operates on the pre-processed audio chunks at a fixed sampling rate of 16 kHz. Drawing upon recent advancements in acoustic emission fault diagnosis for edge computing, we opted for lightweight, highly interpretable statistical descriptors in both the time and frequency domains, which have proven highly effective for tree-based machine learning architectures (Shubita et al., 2023).

For each chunk, five primary statistical features were computed to capture the mechanical acoustic signatures:

- **Root Mean Square (RMS):** Measures the average signal energy, providing a direct correlation to overall vibration and sound intensity.
- **Kurtosis (Pearson, fisher=False):** Quantifies the "peakedness" or impulsiveness of the waveform. Elevated kurtosis values serve as a strong indicator of impulsive faults (e.g., grinding or sudden impacts) in rotating machinery.
- **Crest Factor:** Calculates the ratio of the peak amplitude to the RMS value. High crest factors reveal occasional, sharp acoustic peaks relative to the machine's average energy baseline.
- **Skewness:** Evaluates the asymmetry of the amplitude distribution, which often shifts when a smooth acoustic waveform becomes distorted by a mechanical defect.
- **Spectral Centroid:** Represents the "center of mass" of the magnitude spectrum. This frequency-domain feature summarizes whether the signal's energy is concentrated at lower or higher frequencies, dynamically shifting when fault-related high-frequency components emerge.

The extraction algorithm compiles these features into a structured dataset (features_trackB.csv), appending critical metadata including the chunk filename, the originating recording (parent_file), machine ID, state, and a numeric class label. To ensure data integrity and prevent the artificial inflation of performance metrics, strict deduplication was applied to remove identical overlapping chunks. The finalized, cleaned dataset comprises approximately 1.15 million distinct audio chunks.

Algorithm Selection and Network Architecture

A Random Forest classifier was selected as the primary predictive model. This learning method was prioritized due to its inherent robustness to noisy feature sets, its ability to naturally define non-linear decision boundaries, and its provision of feature importance scores that aid in interpreting mechanical fault contributors.

The forest was configured with an ensemble of 200 trees. Empirical prototyping indicated that this hyperparameter offered an optimal balance between predictive stability and computational cost. Ensembles with fewer than 100 trees exhibited volatile accuracy and recall metrics, whereas increasing the count beyond 200 trees yielded diminishing returns in performance while unnecessarily increasing inference time.

Data Imbalance and Mitigation Strategies

Exploratory analysis of the final dataset revealed a severe class imbalance: approximately 98.8% of the data represented "Normal" operations, while only 1.2% represented "Abnormal" faulty states. To rigorously evaluate the model and prevent data leakage, train-test splitting was executed strictly at the `parent_file` level, ensuring the model never encountered chunks from the same continuous recording across different splits.

Initially, this extreme imbalance induced an accuracy paradox: the model achieved a high overall accuracy (~95%) but exhibited critically low recall on the Abnormal classes. The model optimized its loss by defaulting to the majority class, effectively under-detecting rare faults. To rectify this, two concurrent mitigation strategies were deployed:

1. **Class-Weight Adjustment:** The default loss calculations were replaced with custom `class_weight` dictionaries. By assigning a baseline weight of 1.0 to Normal classes and elevating the Abnormal class weights to a range of 2.0 to 4.0, misclassifications of faulty chunks were heavily penalized. This algorithmic nudging forced the Random Forest to prioritize minority fault patterns.
2. **Targeted Random Oversampling:** To further combat the sparsity of fault data, random oversampling with replacement was applied exclusively to the training set. Abnormal chunks within the training split were resampled until they constituted 20% to 30% of the training volume (up from the baseline 1.2%). Crucially, the validation and test sets remained completely unaltered and severely imbalanced, ensuring that performance evaluation reflected realistic, real-world deployment conditions.

Performance Evaluation and Limitations

The implementation of the 200-tree Random Forest, combined with rigorous parent-group splitting and imbalance mitigation, yielded highly stable metrics. The model sustained an overall accuracy of approximately 95% across both the validation and test sets. For Normal operational classes, the model demonstrated exceptional reliability, achieving precision and recall scores ranging between 0.90 and 0.97.

Regarding the Abnormal classes, the combination of oversampling and class-weight tuning successfully elevated recall compared to the unweighted baseline, particularly for Machine 2. However, the recall for faulty states ultimately plateaued between 0.16 and 0.24, depending on the specific machine. While overall accuracy remains robust and the system successfully identifies a portion of mechanical failures, the extreme sparsity of the original fault data remains the primary limitation of this specific track, highlighting the inherent difficulty of capturing highly rare mechanical anomalies without larger corpuses of faulty acoustic samples.

Experimental Baseline: Statistical Feature Classification

Experimental Design based on Mohammed et al. (2020)

To establish a comparative baseline and validate existing literature, an empirical experiment was conducted replicating the fault diagnosis methodology proposed by Mohammed et al. (2020). In their research, the authors demonstrated that rotating machinery faults could be diagnosed efficiently by extracting specific statistical moments from audio signals. Following their methodology, we extracted eight statistical features across both the time and frequency domains: Mean, Standard Deviation, Skewness, and Kurtosis.

These eight features were fed into a Random Forest classifier to test if this lightweight, mathematically simple approach could sufficiently resolve our highly imbalanced industrial dataset. The experiment was conducted in two phases: an unweighted control run and a class-weighted experimental run.

Phase 1: The Unweighted Control Run (Accuracy Paradox)

In the initial phase, the Random Forest was trained using standard, unweighted loss calculations. The model achieved a seemingly strong **overall accuracy of 81%**.

However, a detailed analysis of the classification report revealed a classic accuracy paradox caused by the dataset's extreme class imbalance.

Because the "Normal" operational data vastly outnumbered the "Abnormal" data, the model optimized its overall accuracy by heavily biasing its predictions toward the majority class (achieving 95%, 80%, and 93% recall for the normal states of Machines 1, 2, and 3, respectively). Consequently, the model failed at its primary predictive maintenance objective detecting faults. The recall scores for the abnormal states were critically low:

- **Class 1 (Machine 1 - Abnormal):** 26% Recall
- **Class 3 (Machine 2 - Abnormal):** 21% Recall
- **Class 5 (Machine 3 - Abnormal):** 25% Recall

This meant the baseline unweighted model was missing approximately 74% to 79% of all actual mechanical faults.

Phase 2: Class Weighting and the Accuracy-Recall Trade-off

To rectify the failure of the control run, the experiment was repeated with algorithmic class weighting (`class_weight='balanced'`) applied to the Random Forest. This mathematical intervention forced the model to penalize misclassifications of the minority abnormal chunks significantly more heavily than the majority normal chunks.

The results of this experimental run perfectly illustrated the Machine Learning Accuracy-Recall trade-off. The **overall accuracy dropped slightly from 81% to 77%**, as the model became more aggressive in predicting faults. However, the recall for the critical abnormal states effectively doubled across the board:

- **Class 1 (Machine 1 - Abnormal):** Improved from 26% to **61%**
- **Class 3 (Machine 2 - Abnormal):** Improved from 21% to **54%**
- **Class 5 (Machine 3 - Abnormal):** Improved from 25% to **49%**

Experimental Conclusion and Feature Importance

From an industrial standpoint, the balanced Phase 2 model is practically superior to the Phase 1 model. In predictive maintenance, the cost of an undetected mechanical failure exponentially outweighs the cost of a premature maintenance inspection.

Furthermore, extracting the Gini importance (feature importance) from the Random Forest validated the core premise of Mohammed et al. (2020). The analysis proved that frequency standard deviation (f_std) and time standard deviation (t_std) were the most powerful discriminators, holding importance weights of approximately 0.26 and 0.24, respectively, followed by time kurtosis (t_kurt at ~ 0.15).

While this baseline experiment successfully proved the diagnostic relevance of these statistical features, its maximum fault detection recall plateaued between 49% and 61%. This performance ceiling explicitly justifies our subsequent transition to the more advanced Convolutional Neural Network (CNN) and Mel-spectrogram architecture, which was strictly necessary to push diagnostic reliability beyond enterprise standards.

Conclusion

The development of an effective machine fault recognition system requires not only advanced classification algorithms but also rigorous, domain-specific data preprocessing. This study successfully demonstrated that transforming raw, inconsistent audio signals into structured data via dynamic trimming, low-pass filtering, and overlapping sliding windows is a vital prerequisite for reliable model training. Furthermore, implementing statistical filtering gates based on the **Coefficient of Variation (CV)** and **Root Mean Square (RMS)** energy proved highly effective in preserving critical mechanical transients while discarding low-quality segments.

The comparative analysis between the Edge Machine Learning baseline and the Deep Learning architecture yielded critical insights regarding the Machine Learning Accuracy-Recall trade-off.

- **Edge ML Limitations:** The baseline experiment, which utilized a 200-tree Random Forest to classify statistical features, validated the diagnostic relevance of metrics like frequency standard deviation and time standard deviation. However, an extreme dataset imbalance (98.8% normal vs. 1.2% abnormal data) severely restricted this lightweight approach. Although class-weight adjustments rectified the initial accuracy paradox and effectively doubled the recall for abnormal states, the fault detection capabilities plateaued at a maximum of 61%.
- **Deep Learning Superiority:** The performance ceiling of the statistical baseline explicitly justified the transition to a Convolutional Neural Network (CNN). The ResNet18 architecture, operating on high-resolution Mel-spectrograms, proved vastly superior at capturing the nuanced, non-linear spatial correlations of mechanical faults. By integrating Focal Loss to dynamically focus on underrepresented anomalies and utilizing a 50% majority voting mechanism for continuous file-level classification, the CNN framework successfully bypassed the limitations of traditional algorithms.

Ultimately, while lightweight statistical models hold value for highly constrained edge deployments, the CNN and Mel-spectrogram architecture is strictly necessary to achieve the rigorous diagnostic reliability demanded by modern industrial predictive maintenance standards.

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